

Application for Managing Todo Lists

Pages 1 to 31: information about the functionality of the application

Pages 32 to 34: information about the source code

Functionality of the Application

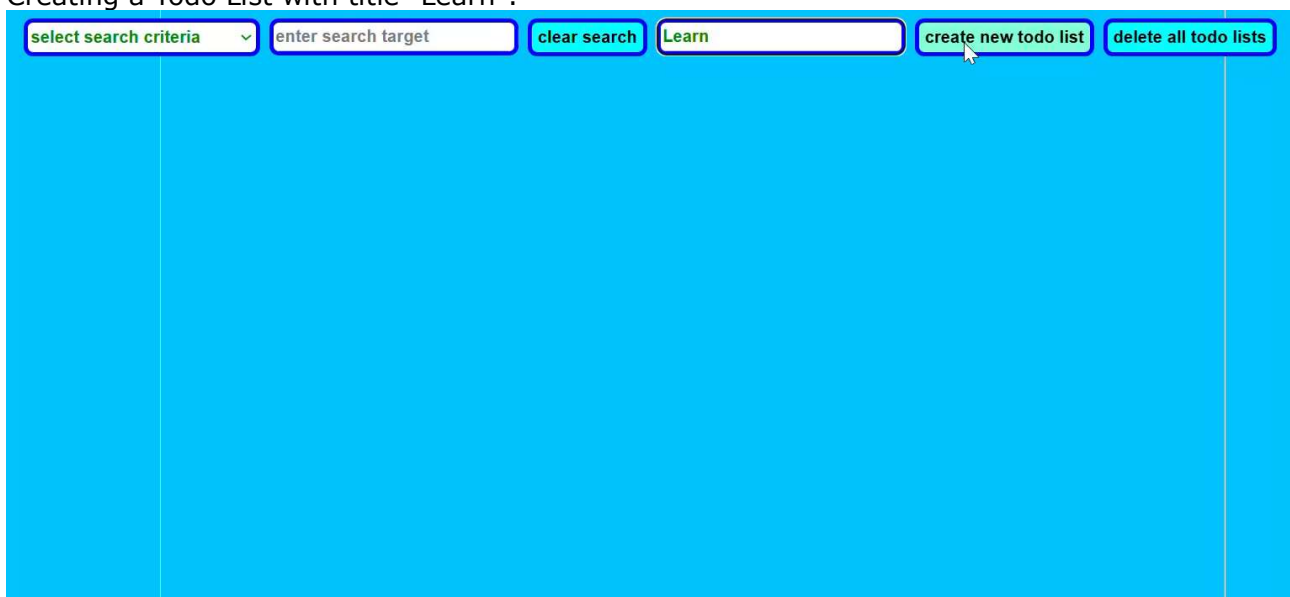
Intuitive and interactive the application for creating and managing many separate Todo Lists, where each Todo List contains its own Todo Tasks. The information about the Todo Lists persists since it is stored in a database, Dexie. The display is always with a live update.

To present visually how the functionalities look like, let's go through a series of screenshots.

The application has started and there are no Todo Lists.



Creating a Todo List with title "Learn".



select search criteria enter search target clear search enter new todo list title create new todo list delete all todo lists				
1	ID 21	Learn	completed 0/0	edit title delete list
1	enter task			add task

The row containing the Todo List Title contains from left to right:

- A button for expanding and folding the list, symbolized with the double arrow.
- Its sequential number among the other Todo Lists.
- ID, which is the ID from the database. The data persists since it is connected to a database.
- Title.
- Ratio of the number of completed tasks to the total number of tasks.
- Button to edit the title.
- Button to delete the entire Todo List.

Add Todo Task "Software Development" to the List "Learn".

select search criteria enter search target clear search enter new todo list title create new todo list delete all todo lists				
1	ID 21	Learn	completed 0/0	edit title delete list
1	Software Development			add task

select search criteria		enter search target	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 18	Learn	completed 0/1	edit title	delete list	
1	Software Development		edit task	delete task		
2	enter task			add task		

The row containing the Todo Task contains from left to right:

- Its sequential number among the other Todo Tasks.
- Description.
- Status symbol "In Progress", symbolized by the "Hammer and Anvil" icon, or "Completed", symbolized by the "Green Circle and Check Mark" icon.
- Button to edit the description.
- Button to delete the Todo Task.

Add Todo Task "Physics" to the List "Learn".

select search criteria		enter search target	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 21	Learn	completed 0/1	edit title	delete list	
1	Software Development		edit task	delete task		
2	Physics			add task		

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 18	Learn	completed 0/2	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	enter task				add task

Add Todo Task "Chemistry" to the List "Learn".

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/2	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry				add task

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 18	Learn	completed 0/3	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	enter task			add task	

Add Todo Task "Everything" to the List "Learn".

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/3	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			add task	

select search criteria

▼

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 18	Learn	completed 0/4	edit title	delete list
1	Software Development	🚩	edit task	delete task	
2	Physics	🚩	edit task	delete task	
3	Chemistry	🚩	edit task	delete task	
4	Everything	🚩	edit task	delete task	
5	enter task			add task	

Creating a second Todo List with title "Buy Healthy Food".

select search criteria

▼

enter search target

clear search

Buy Healthy Food

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development	🚩	edit task	delete task	
2	Physics	🚩	edit task	delete task	
3	Chemistry	🚩	edit task	delete task	
4	Everything	🚩	edit task	delete task	
5	enter task			add task	

select search criteria

▼

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

⌵ 1	ID 18	Learn	completed 0/4	edit title	delete list
1	Software Development	🚩	edit task	delete task	
2	Physics	🚩	edit task	delete task	
3	Chemistry	🚩	edit task	delete task	
4	Everything	🚩	edit task	delete task	
5	enter task			add task	
⌵ 2	ID 19	Buy Healthy Food	completed 0/0	edit title	delete list
1	enter task			add task	

Add Todo Task "Tomatoes 10 kg" to the List "Buy Healthy Food".

select search criteria

▼

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

⌵ 1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development	🚩	edit task	delete task	
2	Physics	🚩	edit task	delete task	
3	Chemistry	🚩	edit task	delete task	
4	Everything	🚩	edit task	delete task	
5	enter task			add task	
⌵ 2	ID 22	Buy Healthy Food	completed 0/0	edit title	delete list
1	Tomatoes 10 kg			add task	

select search criteria

▼

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 18	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task
2	ID 19	Buy Healthy Food	completed 0/1	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	enter task				add task

Add Todo Task "Cabbage 5 pcs" to the List "Buy Healthy Food".

select search criteria

▼

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task
2	ID 22	Buy Healthy Food	completed 0/1	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	Cabbage 5 pcs				add task

select search criteria		enter search target	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 18	Learn	completed 0/4	edit title	delete list	
1	Software Development			edit task	delete task	
2	Physics			edit task	delete task	
3	Chemistry			edit task	delete task	
4	Everything			edit task	delete task	
5	enter task				add task	
2	ID 19	Buy Healthy Food	completed 0/2	edit title	delete list	
1	Tomatoes 10 kg			edit task	delete task	
2	Cabbage 5 pcs			edit task	delete task	
3	enter task				add task	

As mentioned, each Todo List can be expanded or folded by clicking on the double arrow on the top left side of each Todo List Title. By default, when the program is started the Lists are folded.

However, each newly created Todo List is expanded by default, at the time of its creation, so that the field for adding tasks is instantly visible. This was demonstrated above by the creation of the Todo Lists "Learn" and "Buy Healthy Food".

Here is the screenshot, if the application is restarted with the current Todo Lists.

select search criteria		enter search target	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 18	Learn	completed 0/4	edit title	delete list	
2	ID 19	Buy Healthy Food	completed 0/2	edit title	delete list	

Expand Todo List "Learn".

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 18	Learn	completed 0/4	edit title	delete list
click to expand list					
	ID 19	Buy Healthy Food	completed 0/2	edit title	delete list

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 18	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task
2	ID 19	Buy Healthy Food	completed 0/2	edit title	delete list

Fold Todo List "Learn".

select search criteria		enter search target	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 18	Learn	completed 0/4	edit title	delete list	
click to fold list		Software Development		edit task	delete task	
2	Physics			edit task	delete task	
3	Chemistry			edit task	delete task	
4	Everything			edit task	delete task	
5	enter task			add task		
2	ID 19	Buy Healthy Food	completed 0/2	edit title	delete list	

select search criteria		enter search target	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 18	Learn	completed 0/4	edit title	delete list	
2	ID 19	Buy Healthy Food	completed 0/2	edit title	delete list	

The status of each Todo Task can be changed by clicking on the corresponding icon: "Hammer and Anvil" icon for status "In Progress" or "Green Circle and Check Mark" icon for status "Completed". Upon each click on the icons, the status alternates.

Change status of "Physics" in Todo List "Learn" from "In Progress" to "Completed".

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task			add task	

task in progress
click to change status

2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	Cabbage 5 pcs			edit task	delete task

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 18	Learn	completed 1/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task			add task	

2	ID 19	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	Cabbage 5 pcs			edit task	delete task
3	enter task			add task	

Change status of "Physics" in Todo List "Learn" from "Completed" to "In Progress".

select search criteria enter search target clear search enter new todo list title create new todo list delete all todo lists

1		ID 21	Learn	completed 1/4	edit title	delete list
1	Software Development			edit task	delete task	
2	Physics			edit task	delete task	
3	Chemistry			edit task	delete task	
4	Everything			edit task	delete task	
5	enter task					add task

task completed click to change status

2		ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task	
2	Cabbage 5 pcs			edit task	delete task	

select search criteria enter search target clear search enter new todo list title create new todo list delete all todo lists

1		ID 18	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task	
2	Physics			edit task	delete task	
3	Chemistry			edit task	delete task	
4	Everything			edit task	delete task	
5	enter task					add task

2		ID 19	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task	
2	Cabbage 5 pcs			edit task	delete task	
3	enter task					add task

As can be seen, the ratio of number of completed tasks to total number of tasks in the list - on the top right corner of each Todo List - changes instantly upon changes in the status of the tasks.

Each Todo List Title and each Todo Task Description can be edited. When it comes to editing Todo List Title and Todo Task Description as well as creating new Todo Lists and new Todo Tasks, the following rule is observed.

When comparing whether the list title or the task description already exists, the comparison deliberately does not ignore the case, for example, a list title or task description of "a" is considered different from a list title or task description of "A".

Since this is a private and personal todo list, the user might well put different meanings to one and the same text but with different capitalization. Otherwise, if the entered title or task description, after trim(), matches exactly, up to the capitalization, to an already existing list title, respectively task description, the system will not allow it to be created because it already exists.

The comparison for the task description concerns only the Todo List where it is contained. Other Todo Lists might contain the same task description but these are not considered in the comparison since the content of each Todo List stands on its own.

Surely, all these variations can be reworked quickly in whatever way is needed, so the presented variant is just one of many possibilities.

Here is a demonstration, when the entered text is a duplicate or there is no entry in the input field.

Trying to create another Todo List with the title "Learn".

select search criteria		enter search target	clear search	Learn	create new todo list	delete all todo lists
1	ID 21	Learn	completed 0/4	edit title	delete list	
1	Software Development		edit task	delete task		
2	Physics		edit task	delete task		
3	Chemistry		edit task	delete task		
4	Everything		edit task	delete task		
5	enter task			add task		
2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list	
1	Tomatoes 10 kg		edit task	delete task		
2	Cabbage 5 pcs		edit task	delete task		
3	enter task			add task		

select search criteria		enter search target	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 21	Learn	completed 0/4	can not create list title already exists	delete list	
1	Software Development		edit task	delete task		
2	Physics		edit task	delete task		
3	Chemistry		edit task	delete task		
4	Everything		edit task	delete task		
5	enter task			add task		
2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list	
1	Tomatoes 10 kg		edit task	delete task		
2	Cabbage 5 pcs		edit task	delete task		
3	enter task			add task		

Trying to create a Todo List when there is no entry in the input field for the title.

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/0	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task
2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	Cabbage 5 pcs			edit task	delete task
3	enter task				add task

can not create list
title can not be empty

Trying to create another Todo Task "Tomatoes 10 kg" in the List "Buy Healthy Food"

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task
2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	Cabbage 5 pcs			edit task	delete task
3	Tomatoes 10 kg				add task

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task
2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	Cabbage 5 pcs			edit task	delete task
3	enter task				add task

can not add task
task already exists

Trying to create another Todo Task in the List "Buy Healthy Food", when there is no entry in the input field for task description.

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task			add task	
2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	Cabbage 5 pcs			edit task	delete task
3	enter task			add task	

can not add task
task can not be empty

Trying to edit a Todo List Title, where the newly modified title is "Learn".

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task			add task	
2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	Cabbage 5 pcs			edit task	delete task
3	enter task			add task	

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task			add task	
2	ID 22	Learn	completed 0/2	edit title	delete list
		confirm cancel			
1	Tomatoes 10 kg			edit task	delete task
2	Cabbage 5 pcs			edit task	delete task
3	enter task			add task	

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development		edit task	delete task	
2	Physics		edit task	delete task	
3	Chemistry		edit task	delete task	
4	Everything		edit task	delete task	
5	enter task			add task	
2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg	can not modify title title already exists	edit task	delete task	
2	Cabbage 5 pcs		edit task	delete task	
3	enter task			add task	

Trying to edit a Todo List Title when there is no entry in the input field for the title.

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development		edit task	delete task	
2	Physics		edit task	delete task	
3	Chemistry		edit task	delete task	
4	Everything		edit task	delete task	
5	enter task			add task	
2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg		edit task	delete task	
2	Cabbage 5 pcs		edit task	delete task	
3	enter task			add task	

select search criteria
enter search target
clear search
enter new todo list title
create new todo list
delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task
2	ID 22		completed 0/2	edit title	delete list
	enter title	confirm	cancel		
1	Tomatoes 10 kg			edit task	delete task
2	Eco Cabbage 5 pcs			edit task	delete task
3	enter task				add task

select search criteria
enter search target
clear search
enter new todo list title
create new todo list
delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task
2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg	can not modify title title can not be empty		edit task	delete task
2	Cabbage 5 pcs			edit task	delete task
3	enter task				add task

Trying to edit another Todo Task, where the modified task is "Tomatoes 10 kg", in the List "Buy Healthy Food".

select search criteria
enter search target
clear search
enter new todo list title
create new todo list
delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task
2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	Cabbage 5 pcs			edit task	delete task
3	enter task				add task

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task
2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	Tomatoes 10 kg	confirmcancel		edit task	delete task
3	enter task				add task

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task
2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	Cabbage 5 pcs	can not modify task task already exists		edit task	delete task
3	enter task				add task

Trying to edit another Todo Task in the List "Buy Healthy Food", when there is no entry in the input field for task description

select search criteria

▼

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

⌵ 1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development	🚩	edit task	delete task	
2	Physics	🚩	edit task	delete task	
3	Chemistry	🚩	edit task	delete task	
4	Everything	🚩	edit task	delete task	
5	enter task			add task	

⌵ 2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg	🚩	edit task	delete task	
2	Cabbage 5 pcs	🚩	edit task	delete task	
3	enter task			add task	

select search criteria

▼

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

⌵ 1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development	🚩	edit task	delete task	
2	Physics	🚩	edit task	delete task	
3	Chemistry	🚩	edit task	delete task	
4	Everything	🚩	edit task	delete task	
5	enter task			add task	

⌵ 2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg	🚩	edit task	delete task	
2	enter task	confirmcancel	🚩	edit task	delete task
3	enter task			add task	

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task			add task	
2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg	can not modify task task can not be empty		edit task	delete task
2	Cabbage 5 pcs			edit task	delete task
3	enter task			add task	

And here is a demonstration when the editing is successful because the text entry is within the stipulated rules.

Edit Todo List Title "Buy Healthy Food" to "Buy Very Healthy Food"

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task			add task	
2	ID 22	Buy Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	Cabbage 5 pcs			edit task	delete task
3	enter task			add task	

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task
2	ID 22	Buy Very Healthy Food	completed 0/2	edit title	delete list
	confirm cancel				
1	Tomatoes 10 kg			edit task	delete task
2	Eco Cabbage 5 pcs			edit task	delete task
3	enter task				add task

select search criteria

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task
2	ID 22	Buy Very Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	Eco Cabbage 5 pcs			edit task	delete task
3	enter task				add task

Edit Todo Task "Cabbage 5 pcs" to "Eco Cabbage 5 pcs" in Todo List Title "Buy Very Healthy Food"

select search criteria		enter search target	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 21	Learn	completed 0/4	edit title	delete list	
1	Software Development			edit task	delete task	
2	Physics			edit task	delete task	
3	Chemistry			edit task	delete task	
4	Everything			edit task	delete task	
5	enter task				add task	
2	ID 22	Buy Very Healthy Food	completed 0/2	edit title	delete list	
1	Tomatoes 10 kg			edit task	delete task	
2	Cabbage 5 pcs			edit task	delete task	
3	enter task				add task	

select search criteria		enter search target	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 21	Learn	completed 0/4	edit title	delete list	
1	Software Development			edit task	delete task	
2	Physics			edit task	delete task	
3	Chemistry			edit task	delete task	
4	Everything			edit task	delete task	
5	enter task				add task	
2	ID 22	Buy Very Healthy Food	completed 0/2	edit title	delete list	
1	Tomatoes 10 kg			edit task	delete task	
2	Eco Cabbage 5 pcs	confirm	cancel	edit task	delete task	
3	enter task				add task	

select search criteria		enter search target	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 21	Learn	completed 0/4	edit title	delete list	
1	Software Development			edit task	delete task	
2	Physics			edit task	delete task	
3	Chemistry			edit task	delete task	
4	Everything			edit task	delete task	
5	enter task				add task	
2	ID 22	Buy Very Healthy Food	completed 0/2	edit title	delete list	
1	Tomatoes 10 kg			edit task	delete task	
2	Eco Cabbage 5 pcs			edit task	delete task	
3	enter task				add task	

The application has also search functionality and the search can be according to several criteria.

- List ID.
- List Title.
- Ratio of the number of Completed Tasks to the total number of tasks.
- Task Description.

The search is initiated just by typing in the input field for search, provided that there is a selected search criteria. The search is not case sensitive and the search result will always contain all Todo Lists that have the entry, regardless of the capitalization. The person doing the search might know the searched word(s) but might not remember the correct capitalization.

Here is a demonstration.

Hover over the input field for search

select search criteria		enter search target	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 21	Learn	completed 0/4	edit title	delete list	
1	Software Development			edit task	delete task	
2	Physics			edit task	delete task	
3	Chemistry			edit task	delete task	
4	Everything			edit task	delete task	
5	enter task				add task	
2	ID 22	Buy Very Healthy Food	completed 0/2	edit title	delete list	
1	Tomatoes 10 kg			edit task	delete task	
2	Eco Cabbage 5 pcs			edit task	delete task	
3	enter task				add task	

Search by List ID

The search result will return all Todo Lists with IDs that contain the entry. As for example, if there are IDs of "14" and "21" and the input for search is "1", both of these Todo Lists will be returned.

Search by ID "21"

select search criteria

list id

list title

ratio completed tasks, entry format: integer/integer

task description

enter search target

clear search

enter new todo list title

create new todo list

delete all todo lists

		Learn	completed 0/4	edit title	delete list
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task
2	ID 22	Buy Very Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	Eco Cabbage 5 pcs			edit task	delete task
3	enter task				add task

list id

21

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task

Search by ID "2". Both lists have "2" in their IDs, so both lists are in the search result.

list id	2	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task
2	ID 22	Buy Very Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	Eco Cabbage 5 pcs			edit task	delete task
3	enter task				add task

Search by List Title

The search result will return all Todo Lists with titles containing the entry.

As for example, if there are List Titles "Learn" and "Buy Very Healthy Food" and the input for search is "ea", both of these Todo Lists will be returned.

Search by Title "LEARN"

list title	LEARN	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task

Search by Title "EA". Both "Learn" and "Healthy" contain "ea", so both lists are in the search result.

list title	EA	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 21	Learn	completed 0/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task
2	ID 22	Buy Very Healthy Food	completed 0/2	edit title	delete list
1	Tomatoes 10 kg			edit task	delete task
2	Eco Cabbage 5 pcs			edit task	delete task
3	enter task				add task

Search by Ratio of the number of completed tasks to the total number of tasks

The search result will return all Todo Lists with ratios containing the entry.

As for example, if there are Todo Lists with ratio "1/4", "0/1" and "1/45" and the input for search is "1", the search results will return all these Todo Lists.

However, if the search entry is for "1/45" the search result will return only the Todo List with ratio "1/45", provided that there are no other lists that contain such entry, as for example, a ratio "21/4568"

Search by Ratio "2/4"

ratio completed tasks, en	2/4	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 21	Learn	completed 2/4	edit title	delete list
1	Software Development			edit task	delete task
2	Physics			edit task	delete task
3	Chemistry			edit task	delete task
4	Everything			edit task	delete task
5	enter task				add task

Search by Ratio "1/2"

ratio completed tasks, en		1/2	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 22	Buy Very Healthy Food	completed 1/2	edit title	delete list	
1	Tomatoes 10 kg		✓	edit task	delete task	
2	Eco Cabbage 5 pcs		✖	edit task	delete task	
3	enter task				add task	

Search by Ratio that contains "2". Both lists have "2" in their ratio, so both are in the search result.

ratio completed tasks, en		2	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 21	Learn	completed 2/4	edit title	delete list	
1	Software Development		✓	edit task	delete task	
2	Physics		✓	edit task	delete task	
3	Chemistry		✖	edit task	delete task	
4	Everything		✖	edit task	delete task	
5	enter task				add task	
2	ID 22	Buy Very Healthy Food	completed 1/2	edit title	delete list	
1	Tomatoes 10 kg		✓	edit task	delete task	
2	Eco Cabbage 5 pcs		✖	edit task	delete task	
3	enter task				add task	

Search by Task Description

The search result will return all Todo Lists that contain the entry. As for example, if there are different Todo Lists, one having a task "Cabbage 5 pcs" and the other having a task "Study Cabbage Plant" and the search input is "Cabbage", the result will return both of these Todo Lists.

Search by Task containing "study".

task description

study

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 21	Learn	completed 2/5	edit title	delete list
1	Software Development	✓	edit task	delete task	
2	Physics	✓	edit task	delete task	
3	Chemistry	✗	edit task	delete task	
4	Everything	✗	edit task	delete task	
5	Study Cabbage Plant	✗	edit task	delete task	
6	enter task			add task	

Search by Task containing "Eco".

task description

eco

clear search

enter new todo list title

create new todo list

delete all todo lists

1	ID 22	Buy Very Healthy Food	completed 1/2	edit title	delete list
1	Tomatoes 10 kg	✓	edit task	delete task	
2	Eco Cabbage 5 pcs	✗	edit task	delete task	
3	enter task			add task	

Search by Task "cabbage". Both lists have tasks that contain "cabbage", so both lists are in the search result.

task description		cabbage	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 21	Learn	completed 2/5	edit title	delete list	
1	Software Development	✓	edit task	delete task		
2	Physics	✓	edit task	delete task		
3	Chemistry	✖	edit task	delete task		
4	Everything	✖	edit task	delete task		
5	Study Cabbage Plant	✖	edit task	delete task		
6	enter task				add task	
2	ID 22	Buy Very Healthy Food	completed 1/2	edit title	delete list	
1	Tomatoes 10 kg	✓	edit task	delete task		
2	Eco Cabbage 5 pcs	✖	edit task	delete task		
3	enter task				add task	

Of course, all these options for the search results can be quickly reworked in whatever variant it is needed. The presented option is just one of many possibilities.

Live update!

The search results, just as the normal display of all Todo Lists in the database, is with a live update. This also means, for example, that if the search criteria is the list title "Buy Very Healthy Food" and the title is modified to "Buy Healthy Food", while the application is in search mode, the search result will no longer display this Todo List.

Here is how it looks.

Search by Title "Buy Very Healthy Food"

list title		Buy Very Healthy Food	clear search	enter new todo list title	create new todo list	delete all todo lists
1	ID 22	Buy Very Healthy Food	completed 1/2	edit title	delete list	
1	Tomatoes 10 kg	✓	edit task	delete task		
2	Eco Cabbage 5 pcs	✖	edit task	delete task		
3	enter task				add task	

Modify Title "Buy Very Healthy Food" to "Buy Healthy Food", while in search mode.

list title	▼	Buy Very Healthy Food	clear search	enter new todo list title	create new todo list	delete all todo lists
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1	ID 22	Buy Healthy Food	completed 1/2	edit title	delete list
		confirm	cancel		
1	Tomatoes 10 kg	✓	edit task	delete task	
2	Eco Cabbage 5 pcs	✗	edit task	delete task	
3	enter task			add task	

The modified list is filtered out of the search result since it no longer contains the search entry.

list title	▼	Buy Very Healthy Food	clear search	enter new todo list title	create new todo list	delete all todo lists
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Source Code

Database: Dexie

Frameworks & Tools: React, Vite, TypeScript, HTML, CSS

Project Structure Overview

```

└─ todo-application
  ├── dist
  ├── node_modules
  ├── public
  └── src
      ├── components
      │   ├── tableTodo
      │   │   ├── popoversTableTodo
      │   │   │   ├── PopoverEditTaskOrDisplayExistingTask.tsx
      │   │   │   ├── PopoverEditTitleOrDisplayExistingTitle.tsx
      │   │   │   ├── PopoverMapHoverListIsExpandedStatusDisplay.tsx
      │   │   │   ├── PopoverMapHoverTaskStatusDisplay.tsx
      │   │   │   └── PopoverMapIdToNewTaskNotValidDisplay.tsx
      │   │   ├── TableBody.tsx
      │   │   ├── TableFooter.tsx
      │   │   ├── TableHeader.tsx
      │   │   └── TableWrapper.tsx
      │   ├── topPanel
      │   │   ├── popoversTopPanel
      │   │   │   ├── PopoverDisplaySearchInstructions.tsx
      │   │   │   └── PopoverTitleNotValidDisplay.tsx
      │   │   └── TopPanel.tsx
      │   └── ProcessTodo.tsx
      ├── data
      │   └── TS db.ts
      ├── styles
      │   ├── # database.css
      │   └── # displayTodo.css
      ├── util
      │   ├── dataStructures.tsx
      │   └── functions.tsx
      ├── App.tsx
      ├── # index.css
      ├── main.tsx
      ├── .gitignore
      ├── eslint.config.js
      ├── index.html
      ├── package-lock.json
      ├── package.json
      ├── README.md
      ├── tsconfig.app.json
      ├── tsconfig.json
      ├── tsconfig.node.json
      └── vite.config.ts
```

The names of the folders, files, methods, fields, etc., are self-explanatory and where needed there are comments.

Main folders is src: [components](#) | [data](#) | [styles](#) | [util](#)

[components/ProcessTodo.tsx](#)

The entry point of the application that is supplied to App.tsx.
This file puts together the work done by all the other files.

[components/tableTodo/TableHeader.tsx](#)

Processes the title of each todo list.

[components/tableTodo/TableBody.tsx](#)

Processes the todo tasks of each todo list.

[components/tableTodo/TableFooter.tsx](#)

Processes the creation of new todo tasks of each todo list.

[components/tableTodo/TableWrapper.tsx](#)

This file provides a framework for the Header, Body and Footer of the table
and it puts together the work done in these files.

[components/tableTodo/popoversTableTodo/PopoverEditTaskOrDisplayExistingTask.tsx](#)

Processes the edit of the tasks. If there is no edit, it displays the existing tasks.

[components/tableTodo/popoversTableTodo/PopoverEditTitleOrDisplayExistingTitle.tsx](#)

Processes the edit of the title. If there is no edit, it displays the existing titles.

[components/tableTodo/popoversTableTodo/PopoverMapHoverListIsExpandedStatusDisplay.tsx](#)

Processes the expanding and folding of the lists.

[components/tableTodo/popoversTableTodo/PopoverMapHoverTaskStatusDisplay.tsx](#)

Processes the status of the tasks in each list.

[components/tableTodo/popoversTableTodo/PopoverMapIdToNewTaskNotValidDisplay.tsx](#)

Processes the validity of the entries for new tasks. The entry must not be empty and
it must not be a duplicate of another task, case sensitive, in the corresponding list.

[components/topPanel/TopPanel.tsx](#)

Processes the search functionality, the creation of new lists and the deletion of all lists.

[components/topPanel/PopoverDisplaySearchInstructions.tsx](#)

Processes the display of search instructions upon hovering in the search input field.

[components/topPanel/PopoverTitleNotValidDisplay.tsx](#)

Processes the validity of the entries for new titles. The entry must not be empty and
it must not be a duplicate of another title, case sensitive, in the database.

[data/db.ts](#)

Creates the Dexie database.

[styles/database.css](#)

Contains the styles shared across most of the files

[styles/displayTodo.css](#)

Contains the styles shared mostly by the files directly processing the table.

[util/dataStructures.tsx](#)

Contains data structures shared across many files of the application.
These data structures are imported in the corresponding files as typed imports.

[util/functions.tsx](#)

Contains helper functions that are applied across the files of the application.